Reading Group 20/03/2024

Introduction

Background - Structural Causal Model

- An SCM $\mathcal{M}$ is a 5-tuple $\mathcal{M} = (\mathcal{V}, \mathcal{N}, \mathcal{U}, \mathcal{F}, P(.))$
- Each observed variable $V_i \in \mathcal{V}$ is realized as an evaluation of the function $f_i \in \mathcal{F}$ looks at a subset of the remaining observed variables
  - Remaining observed variables $Pa_i \subset \mathcal{V}$ (Parent Set of V_i)
  - Unobserved exogenous noise variable $E_i \in \mathcal{N}$
  - Unobserved confounding variable $U_i \in \mathcal{U}$
- $P(.)$ is a product joint distribution over all unobserved variables $\mathcal{N} \cup \mathcal{U}$

Background - SCMs and Causal Graphs

- Each SCM induces a directed graph called the causal graph or acyclic directed mixed graph (ADMG)
- In the context of this paper, the graph is still acyclic, but no longer a DAG
  - The graph can handle bidirectional edges (e.g. $V_i \leftrightarrow V_j$) to represent unobserved confounders
- V_i is called an ancestor of V_j if there is a directed path from V_i to V_j
- A subset $g \subset G$ is called a c-component if it is a maximal set of nodes in G that are connected by bi-directed paths

Introduction

- Goal: Perform Causal Inference, e.g. evaluating the causal effect of an intervention on a system of interest
- Today, most of our datasets contain high-dimensional variables, such as images
- Modern deep learning architectures can handle high dimensional data and solve ML problems such as prediction or learning distributions
- However, they are not designed to perform causal inference
  - They fail to generalize where there exists spurious correlations in the data

Main Example of the paper

- Healthcare dataset with Symptoms, Diagnosis and XrayImages of a collection of patients
  - Contains a mixture data from two hospitals in different cities where one of the cities is more wealthy, where patients are less likely to fall sick on average
- Hospital location acts as a latent confounder that affect both the Symptoms and Diagnosis

Main Example of the paper

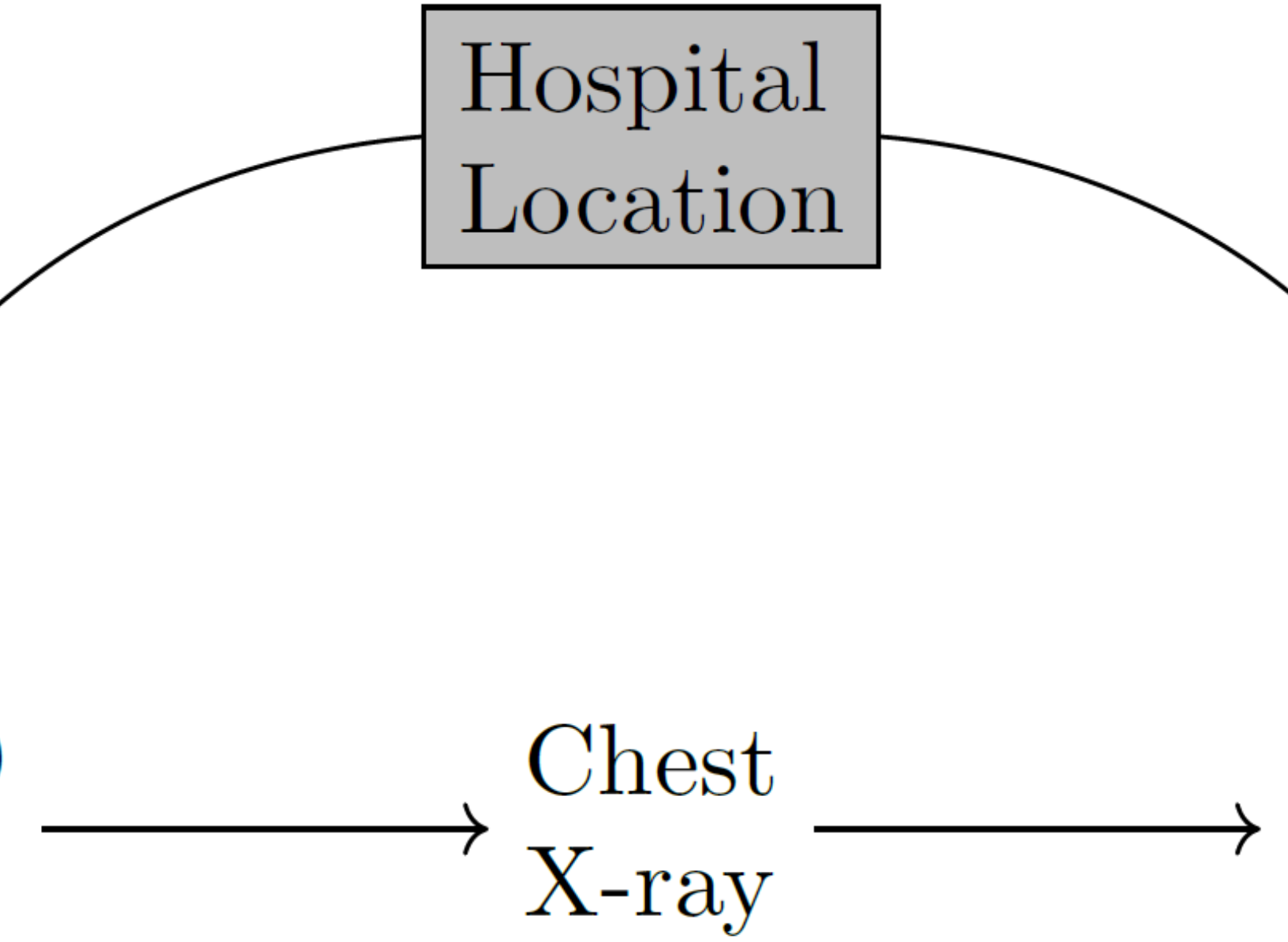


Figure 1: Causal Graph of the main example

Main Idea of the paper

- The paper proposes a modular sampling-based framework for deep causal generative models

- Collect observational samples and use these samples to estimate the interventional distribution
- In this paper, $\mathcal{L}_1$, $\mathcal{L}_2$, and $\mathcal{L}_3$ distributions as notation for observational, interventional, and counterfactual distributions respectively
- Using deep learning architectures that mimic the causal structure of the system
- Using pre-trained generative models like DALL-E, ChatGPT or Bard

Main Idea of the paper

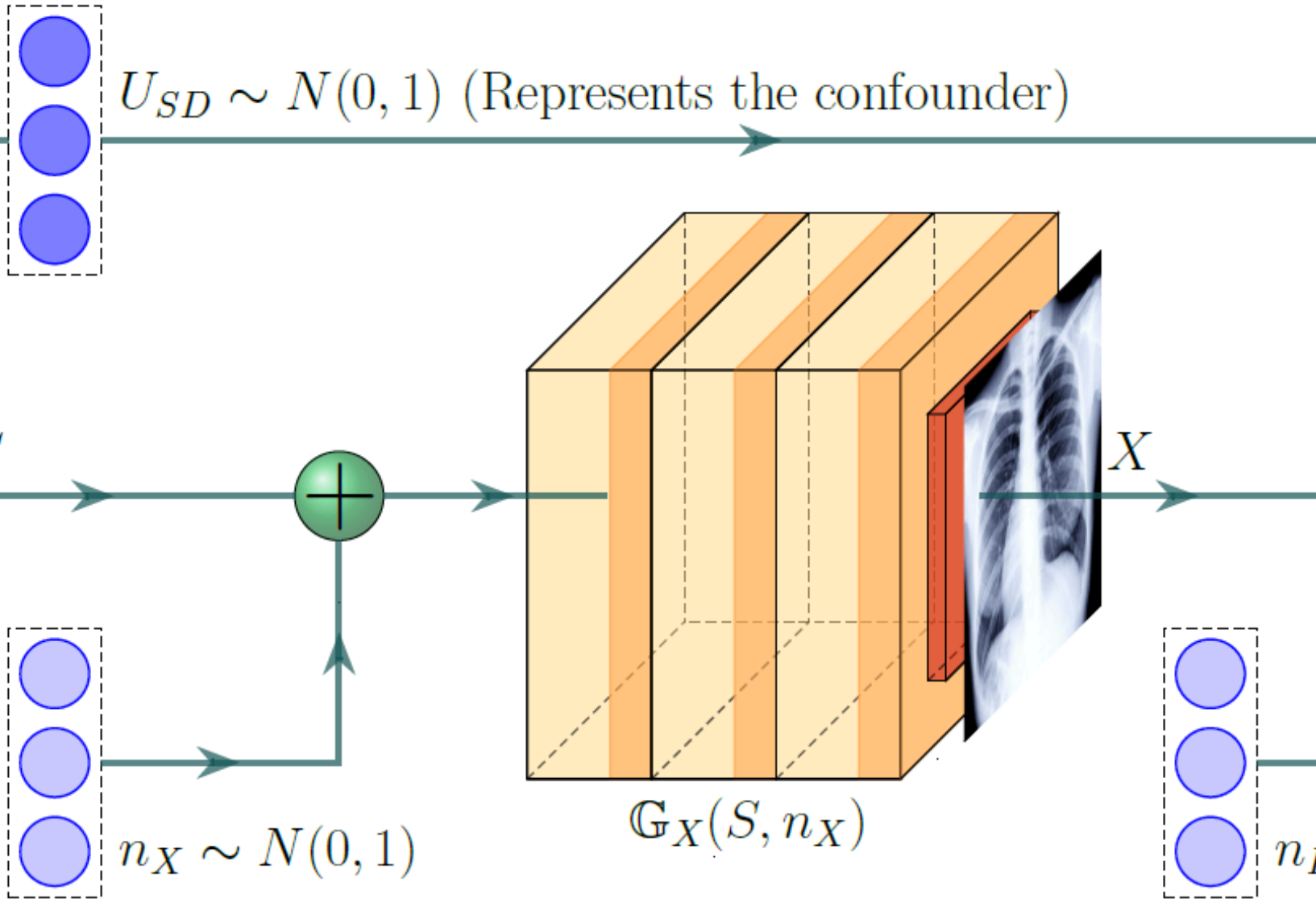


Figure 2: Deep Causal Generative Model

Deep Causal Generative Models with Unobserved Confounders

Definition

Let a SCM $\mathcal{M}$ be made up of functions $X_i = f_i(Pa_i, E_i)$.

A neural net architecture $\mathbb{G}$ is called a deep causal generative model (DCM) for an ADMG G inducing a distribution Q if it is composed of a collection of neural nets, one $\mathbb{G}_i$ for each $V_i \in \mathcal{V}$ such that

- Each $\mathbb{G}_i$ accepts a sufficiently high-dimensional noise vector N_i
- The output of $\mathbb{G}_j$ is input to $\mathbb{G}_i \Leftrightarrow V_j \in Pa_G(V_i)$
- $N_i = N_j \Leftrightarrow V_i \leftrightarrow V_j$

Note that $\mathbb{G}$ will contain **one neural net per observed variable**: $\mathbb{G} = \left\{\mathbb{G}_{Z_1}, \mathbb{G}_{Z_2}, ...\right\}$

Intuition

- Let $D \sim P(\mathcal{V})$ be an observational sample
- Train a DCM $\mathbb{G}$ that fits the observational distribution $P(\mathcal{V})$ to sample the interventional distribution $\mathcal{L}_2$
- Training involves identifying which causal mechanisms should be trained together based on their dependencies in the causal graph
- Start with the first neural network $\mathbb{G}_{Z_1} \in \mathbb{G}$ to learn a mapping to sample $P(Z_1 \mid Z_3, X_1)$
- The training preserve the integrity of c-components and factorize the joint distribution of them to yield $P(\mathcal{V})$

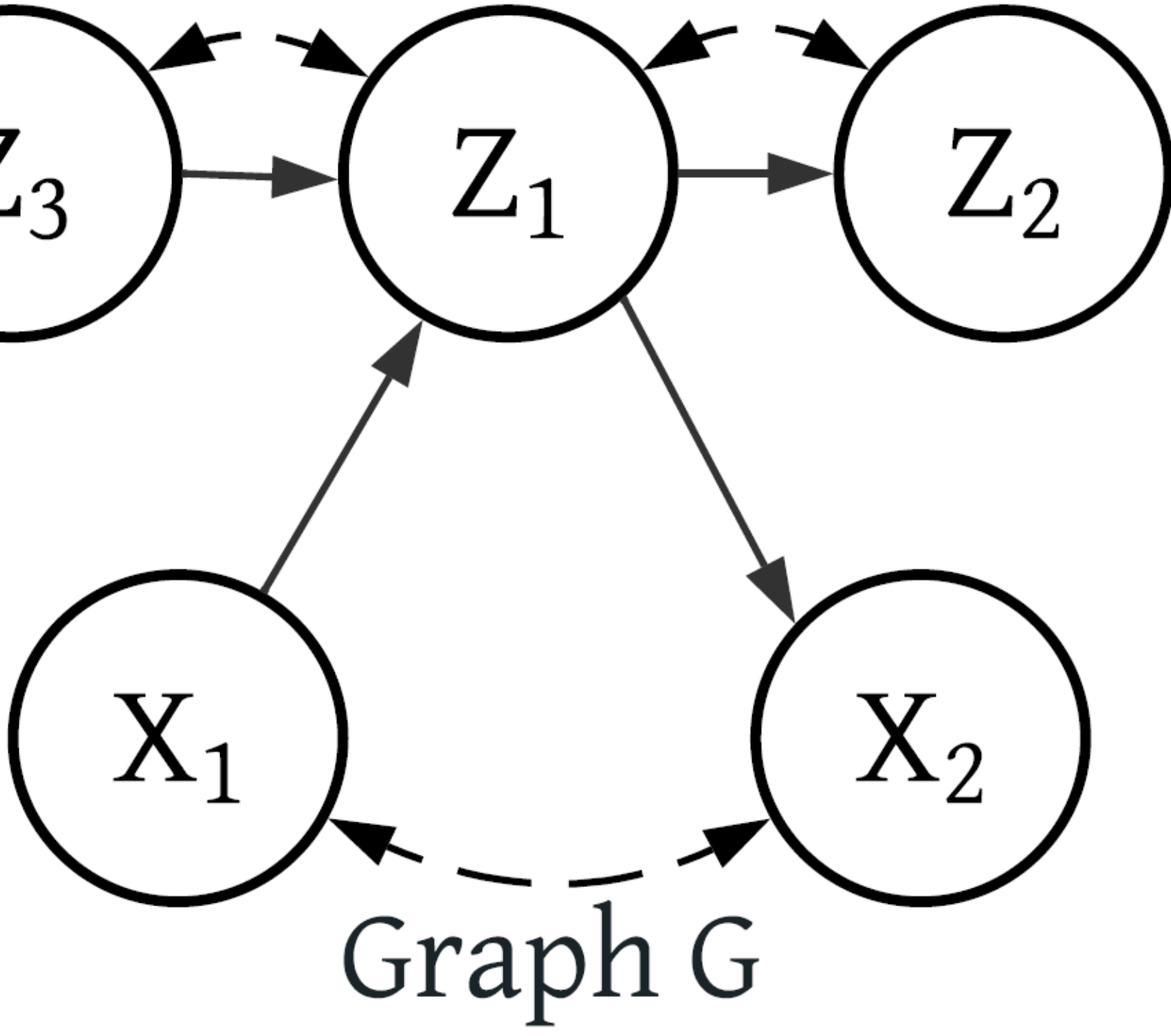


Figure 3: Example Causal Graph: X denotes the first c-component, Z the second c-component

Intuition

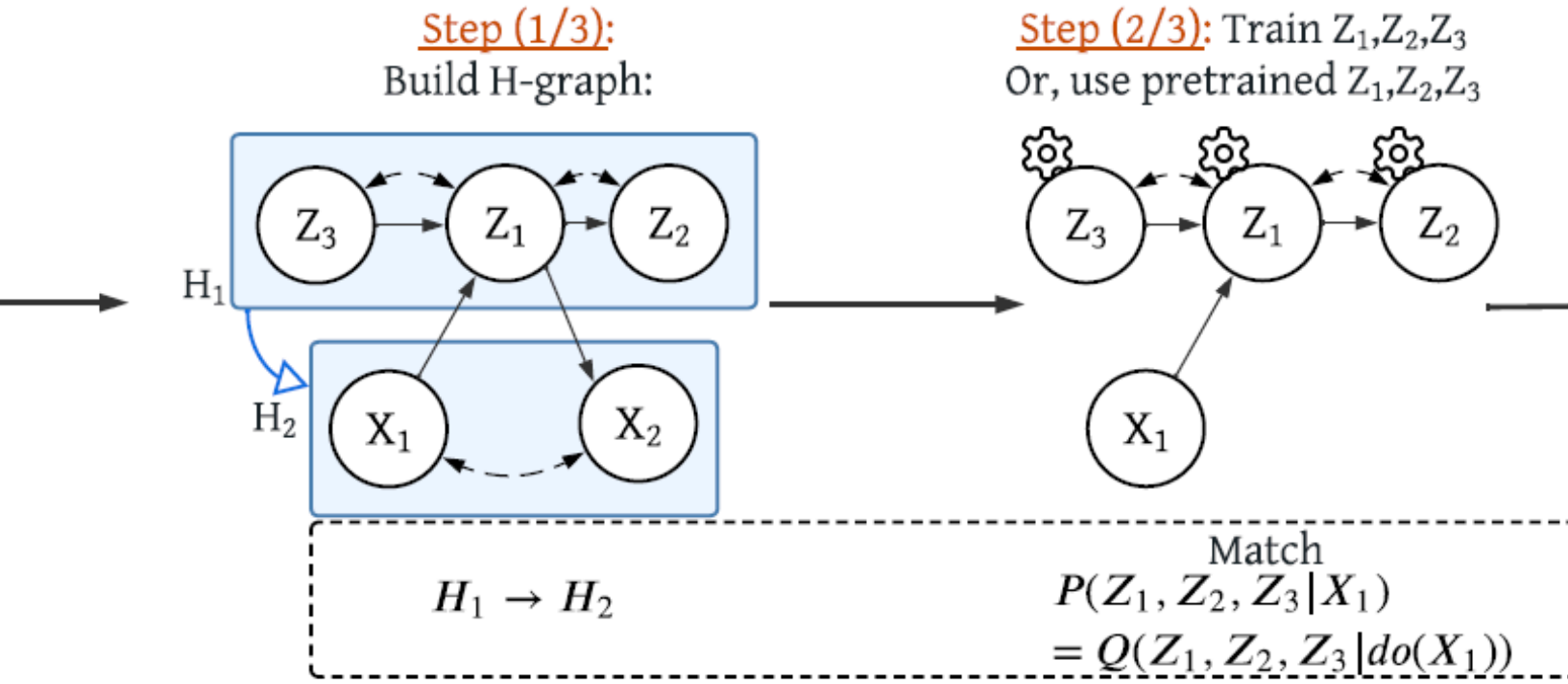


Figure 4: Modular training in 3 steps. Step 1: Since $P(X_1, X_2 \mid do(Z_1)) \neq P(X_1, X_2 \mid Z_1)$, add an edge from c-component $H_1 : [Z_1, Z_2, Z_3]$ to c-component $H_2 : [X_1, X_2]$. Step 2: Train only the mechanisms in H_1 to match $P(Z_1, Z_2, Z_3 \mid X_1) = Q(Z_1, Z_2, Z_3 \mid do(X_1))$. Step 3: Train only H_2 while using pre-trained Z_1, Z_3 to match $P(X_1, X_2, Z_1, Z_3) = Q(X_1, X_2, Z_1, Z_3)$

Intuition

- By training mechanisms in the same c-component together, the complexity of matching joint distributions involving high and low-dimensional variables is significantly reduced.
- Modularizing the training process allows for more efficient training and simplifies the overall learning task.

Experiments

Performance on Real-world COVIDx CXR-3

- 30000 XRay images with labels C (Covid) and N (Pneumonia)
  - Healthy patiens: $C = 0, N = 0$
  - Non-Covid Pneumonia patiens: $C = 0, N = 1$
  - Covid Pneumonia patiens: $C = 1, N = 1$
- Patient's location is a hidden confounder

- A persons socio-economic and health conditions in a specific location might affect both the likelihood of getting Covid and being properly diagnosed with Pneumonia by local health care

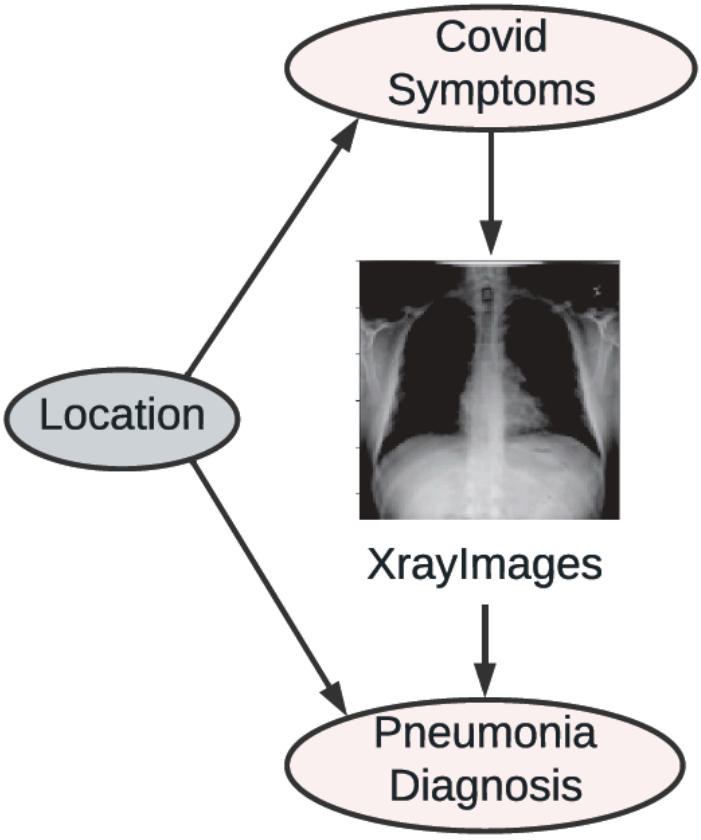


Figure 5: COVIDx CXR-3 graph

Performance on Real-world COVIDx CXR-3

- Fit the joint distribution $P(XRay, C, N)$
- Modular training of $\left[\mathbb{G}_{Xray}\right] \rightarrow [\mathbb{G}_C, \mathbb{G}_N]$
  - Using pre-trained generative model for X-Ray images $\mathbb{G}_{Xray}$
  - Train $\mathbb{G}_C$ and $\mathbb{G}_N$ together as they are in the same c-component
- The joint distribution contains both low and high-dimensional variables
  - Map Xray to a low-dimensional representation Rxray with an encoder and match $P(C, Rxray, N)$

Performance on Real-world COVIDx CXR-3

- The score which is minimized is the Total Variation Distance (TVD)
- The average treatment effect, i.e., the difference between $E[P(N \mid do(C = 1))]$ and $E[P(N \mid do(C = 0))]$ is in $[0.05, 0.08]$

- Intervention with Covid increases the likelihood of being diagnosed with Pneumonia

Conclusion

Conclusion

- The paper propose Modular-DCM for learning deep causal generative models for estimating causal effects with high-dimensional variables in the presence of unobserved confounders
- Modular-DCM can generate high-dimensional samples from identifiable interventional and counterfactual distributions
- Main Limitation: The paper assumes that the causal graph is fully specified with latents and the causal model is semi-Markovian (the probability of there being a change in the hidden state depends on the amount of time that has elapsed since entry into the current state (?))

Bibliography